

VAISHNAVI SINGH

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SUMMARY

Computer Engineering undergraduate (B.Tech, 2028) with strong Data Structures and Algorithms skills (140+ LeetCode problems, 1540 rating) and full-stack experience building custom front-end and back-end applications with Java, the MERN stack, Python/Flask and SQL. Skilled in RESTful API development, web services, database design, third-party integration, testing and troubleshooting, and building scalable internal tools that automate business processes. Strong communicator who translates business requirements into technical solutions. Seeking Application Engineering internship roles.

EDUCATION

Thapar Institute of Engineering and Technology, Patiala, India

Aug 2024 - May 2028

Bachelor of Technology (B.Tech), Computer Engineering | CGPA: 8.82 / 10

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Software Engineering

TECHNICAL SKILLS

- **Programming Languages:** C++, Java, Python, JavaScript (ES6+), TypeScript, SQL, C
- **Web Development:** React.js, Node.js, Express.js, Flask, HTML5, CSS3, RESTful Web Services, API Development, Microservices, JWT Authentication, Role-Based Access Control
- **Databases:** MySQL, MongoDB, Mongoose, Schema Design, Query Optimization
- **Data & Machine Learning:** Pandas, NumPy, Scikit-learn, Matplotlib, Feature Engineering, Logistic Regression, TF-IDF, Model Evaluation
- **Computer Science Fundamentals:** Data Structures, Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Problem Solving
- **Tools & Practices:** Git, GitHub, GitHub Actions (CI/CD), Postman, VS Code, Jupyter, Jest, Unit Testing, Technical Design, Test Documentation, Debugging, Agile/Scrum

PROJECTS

OpsDesk | Internal Workflow Automation & Request Triage Portal

React.js, TypeScript, Node.js, Express.js, MongoDB, MySQL, Python (Flask), Scikit-learn, JWT

- Built an internal business application that automates IT, procurement and leave request workflows with multi-level approvals and role-based access control (JWT) for employee, manager and admin roles, replacing spreadsheet and email tracking.
- Developed RESTful APIs in Node.js/Express with MongoDB for request data and designed a normalized MySQL schema for audit logs and SLA reporting, writing SQL queries with joins, aggregations and indexes for an analytics dashboard.
- Trained a TF-IDF and Logistic Regression text classifier on approximately 2,000 labelled tickets to auto-categorize and route incoming requests to the right team with approximately 85% accuracy, served through a Flask microservice.
- Integrated third-party services (Google Sheets API, Nodemailer email alerts), wrote Jest unit tests with test documentation, and deployed on Render with CI/CD via GitHub Actions.

RewardMax | Smart Rewards & Cashback Optimizer

React.js, Node.js, Express.js, MongoDB, Python (Flask), REST APIs

- Engineered a formula-based rewards-optimization engine across 12 reward categories, ranking offers by reward rate, monthly cap and logged spend, cutting manual comparison time by approximately 35%.
- Built the Node.js/Express REST API backend with MongoDB and a Python (Flask) microservice, sustaining approximately 150 requests per second at approximately 120 ms average latency; built a React.js spend dashboard.

ChurnGuard | Customer Loyalty & Churn Risk Dashboard

React.js, Node.js, Express.js, MongoDB, Python, Scikit-learn, Pandas, NumPy

- Built an RFM-based feature-engineering pipeline in Pandas and NumPy over approximately 5,000 customer records and trained a Logistic Regression classifier achieving 87% accuracy and a 0.84 F1-score.
- Implemented low, medium and high risk banding that flagged approximately 18% of customers as high-risk across 900+ accounts to help relationship managers prioritize outreach, exposed through an Express.js REST API and a React.js monitoring dashboard.

PROFESSIONAL EXPERIENCE

Student Placement Representative, Thapar Institute

Aug 2025 - Present

- Coordinated between the placement cell and visiting companies, streamlining communication on drive schedules and registrations.

Web Team Member, Microsoft Learn Student Chapter (MLSC), Thapar Institute

Sep 2024 - Mar 2025

- Developed the About Us and Timeline pages for Makethon-7 using HTML, CSS and JavaScript for 300+ visitors.

Proctor, Vahini Hall, Thapar Institute

Aug 2025 - Present

- Responsible for student welfare, discipline and administration for a resident community of 900+ students.

Coordinator, Student Alumni Interaction Cell, Thapar Institute

Sep 2024 - Mar 2025

- Organized alumni-student outreach events for 200+ students, handling scheduling, logistics and communication.

ACHIEVEMENTS & AWARDS

- Semifinalist, Flipkart GRID 8.0 Software Development Challenge (2026); Best Speaker, Model United Nations Society (2026).
- Gold Medalist, Inter-University Girls' Badminton, PEC Chandigarh (Oct 2025); 2nd Prize, Thapar Quizzing Club GK Quiz (Feb 2025); 3rd Prize, Construct, Thapar Civil Society (Oct 2024).